WK 01: Big Data

A New Paradigm

PPOL 5206 | Massive Data Fundamentals | Spring 2026
January 8th, 2026

Questions we can now Answer

The Power of Massive Data

The Policy Promise

We can now answer questions that we could only ask before.

What's now possible:

- **Fraud detection at scale:** CMS recovers billions annually from Medicare/Medicaid fraud using pattern detection across millions of claims
- **Real-time public health surveillance:** COVID wastewater monitoring, flu tracking from search queries, syndromic surveillance
- **Climate and environmental modeling:** Satellite imagery, sensor networks, atmospheric data at planetary scale
- **Urban operations:** 311 service optimization, traffic flow, predictive infrastructure maintenance
- **Census and survey operations:** Disclosure avoidance, differential privacy, administrative data linkage

The Dark Side

The same capabilities raise serious policy questions:

- Algorithmic bias in
    - criminal justice
    - Lending
    - hiring
- Surveillance and consent
- Privacy vs. utility tradeoffs

These are a few of the concerns that the prevalence of big data raises.

Data as Public Infrastructure

Data infrastructure increasingly resembles physical infrastructure:

- Roads, utilities, postal system → data systems, APIs, platforms

Policy questions:

- Who should build and maintain it?
- Who should have access?
- What are the network effects?
- What happens when it's privately controlled?

Data Sovereignty

Data is becoming a geopolitical issue:

- **GDPR (EU):** Privacy rules with extraterritorial reach
- **Data localization laws:** Requirements to store data within national borders
- **AI training data:** Strategic competition over datasets

Data flows are trade policy, foreign policy, and security policy.

Three Persistent Tradeoffs

Privacy vs. Utility

- More data linkage enables better analysis—and creates privacy risks
- No free lunch: privacy protection costs statistical precision

Interpretability vs. Performance

- The most accurate models are often the least explainable
- Policy requires justification—can you explain it to a judge? A legislator? The public?

Speed vs. Rigor

- Operations demand quick answers; research demands careful ones
- Different questions require different standards

Learning Objectives

By semester's end, you will be able to:

Technical Foundations

- Query relational databases using SQL
- Process datasets that exceed single-machine capacity using distributed computing tools
- Navigate cloud platforms to provision and manage scalable data infrastructure
- Use version control and command-line tools that support collaborative data work

Analytical Judgment

- Identify computational, storage, and cost constraints when evaluating data solutions
- Compare architectural approaches (data warehouses, lakes, lakehouses) for policy applications
- Evaluate trade-offs across major cloud platforms for organizational needs

Learning Objectives

By semester's end, you will be able to:

Governance and Context

- Analyze data regulations affecting collection, storage, and cross-border transfer
- Apply ethical frameworks to data-intensive policy decisions

Synthesis

- Design and justify a data architecture recommendation for a policy use case

What is Data?

Foundations

What is data?

- Text
- Dates & Times
- Numbers
- Audio
- Images
- Sensor readings
- Etc…

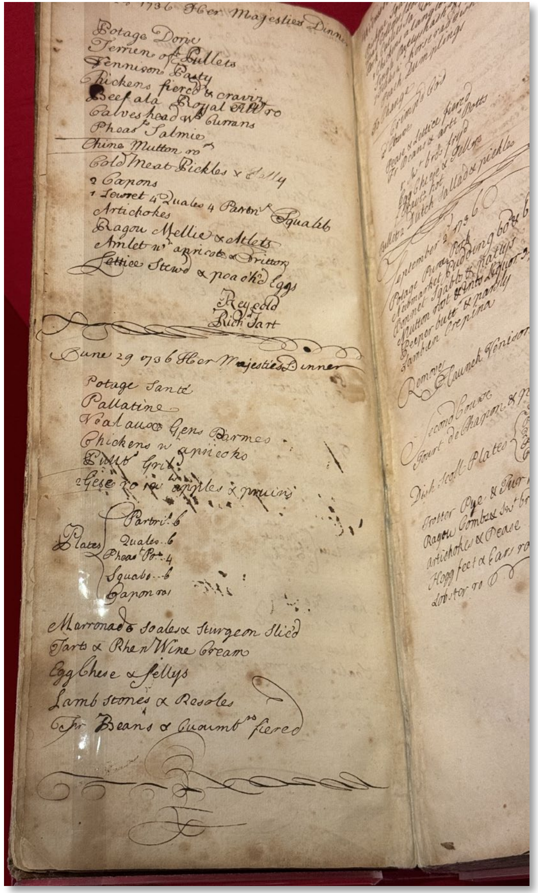


Image: Dinner log from Kensington Palace Archives.

Data has always existed.

What changes is our capacity to capture, store, and process it.

Image: Time series of a dinner service from Kensington Palace Archives

The Long History of Data Challenges

Historical Perspectives

Data: A Human Urge

Ishango Bone (~20,000 BCE):

- Possibly the oldest known record of numerical data
- Tally marks carved into bone
- Recording information is a fundamental human behavior

Constraint:

- High cost of materials (bone, clay, papyrus, vellum)
- Societies had to be selective about what was worth recording

Source: https://www.naturalsciences.be/en/museum/exhibitions-activities/exhibitions/250-years-of-natural-sciences/the-ishango-bone

The First Recorded Data

Sumerian cuneiform tablets (~3400 BCE):

- Accounting and inventory records
- The world's first accountants

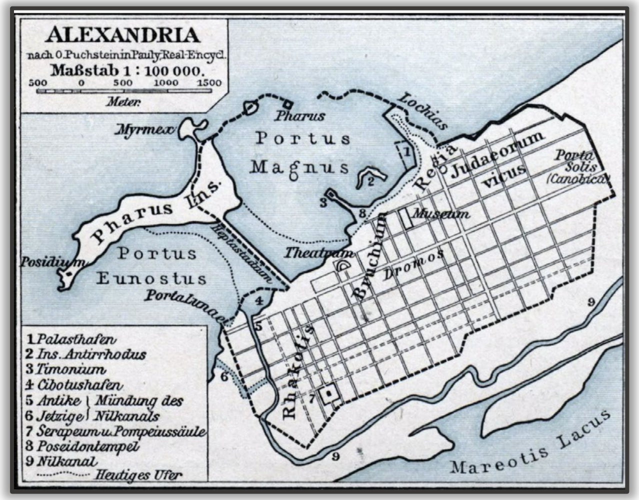
Constraints:

- Limited literacy *(read)*
- Manual recording *(write)*
- Physical storage *(store)*
- Heavy tablets *(Transmission)*

Source: [How the world's first accountants counted on cuneiform - BBC News]

Centralized Knowledge

Library of Alexandria: First attempt to centralize recorded knowledge
Advantage: Data gains value when compared with other data and more easily retrieved
Constraint: Copying data was labor and skill intensive work



Source: [History of the Ancient Library of Alexandria - Historic Mysteries]

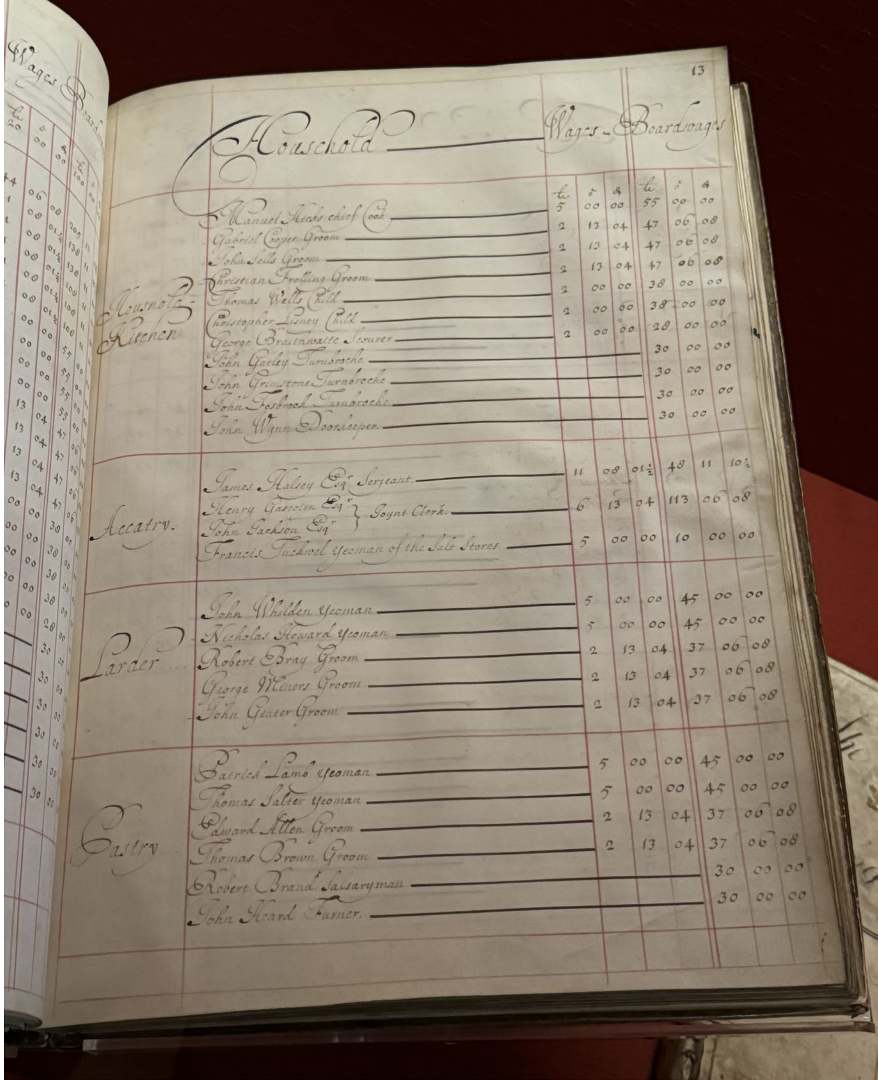
Organizing Data

Double Entry Bookkeeping in Italy (Late 1200's – 1500's)

- Error checking
- Analysis (some)
- Systematic financial records

Constraints:

- Manual copying
- No broad standardization
- Fragile media
- Limited I/O



Source: Account log book from Kensington Palace

Analyzing Systems Too Large to Track

- “Big Data” problems hit the physics community in the 19th century
    - gases contain ~10^{23} molecules
    - Tracking individual particles was impossible
    - Newtonian determinism broke down

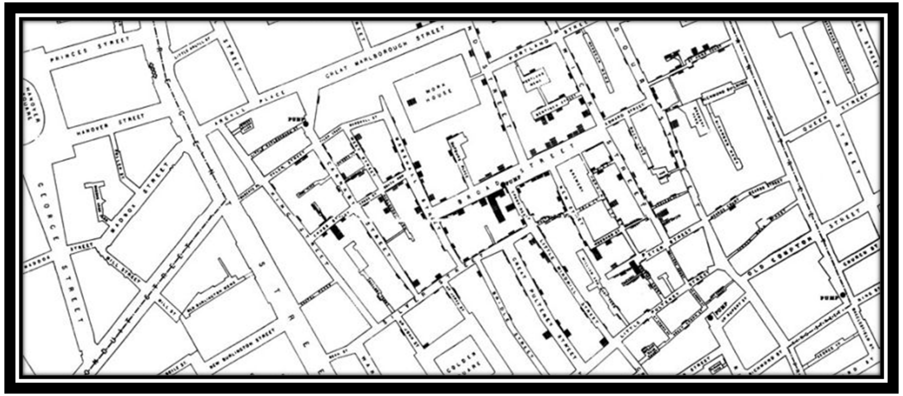
A new paradigm was required

- The solution: statistical mechanics (Maxwell, Boltzmann, Gibbs)
    - Describe aggregate behavior through probability distributions
    - Accept that individual-level precision is neither possible nor necessary
    - Emergent patterns become the answer

Key insight: When you can't analyze everything, you change what "analysis" means.

Visualization as Analysis

- 1854 London cholera outbreak: 616 deaths in Soho
- John Snow mapped deaths by location
- Pattern emerged: deaths clustered around the Broad Street pump
- The map made visible what tables could not show
- Result: pump handle removed, outbreak contained



Constraint addressed: Human cognition - we see spatial patterns that we miss in rows and columns

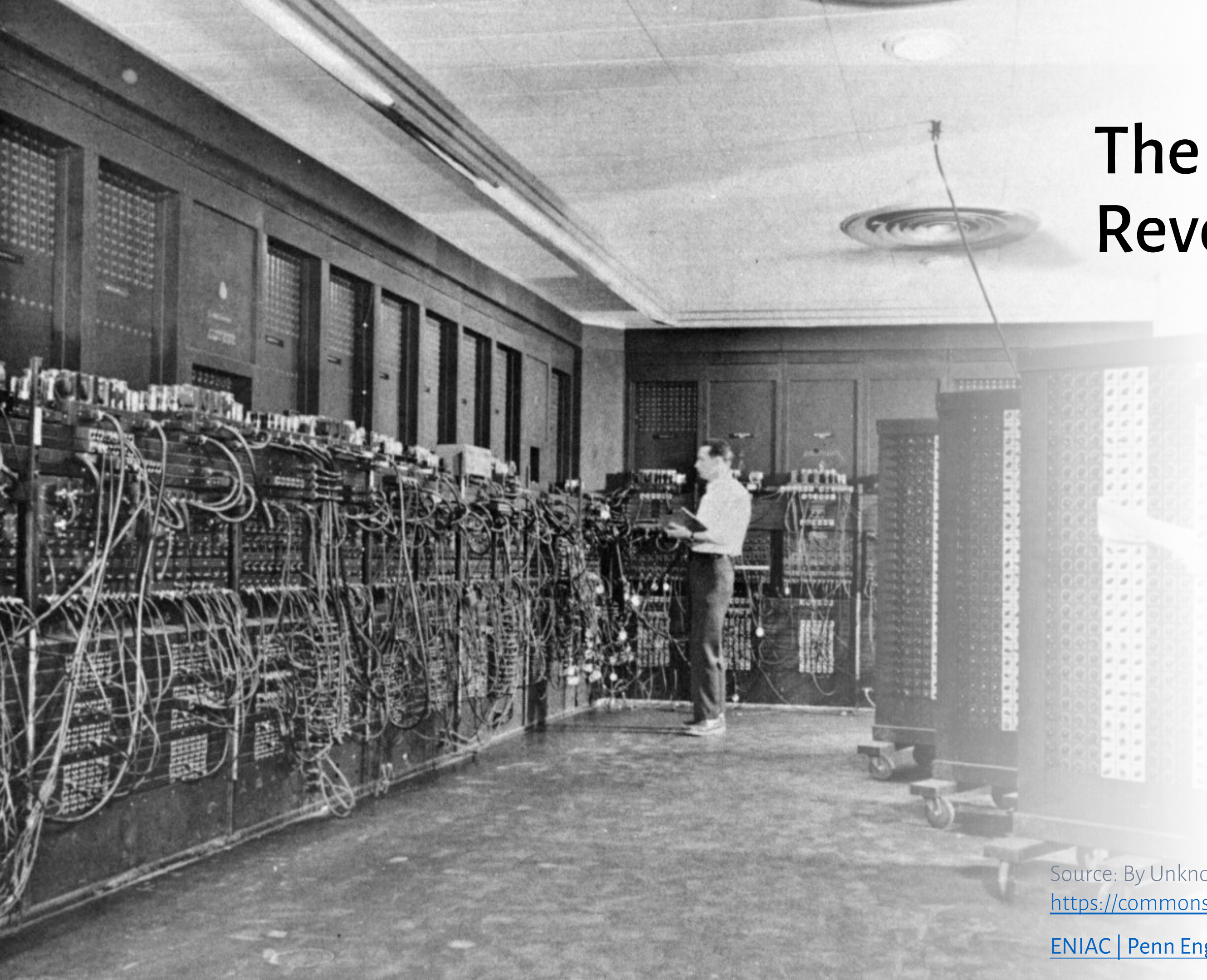
Source: Snow, J. (1855). On the Mode of Communication of Cholera. London.

The Broad Street Pump (2024)

The Tabulation Crisis

- Following the 1880 census, the Census Bureau was collecting more data than it could tabulate
  - 1888 competition to find a more efficient method
- Three contestants accepted the challenge:
  - The first two captured the data in 144.5 hours and 100.5 hours
  - The third, a former Census Bureau employee named Herman Hollerith, completed the data capture process in 72.5 hours
- Next, the contestants had to prove they could prepare data for tabulation (age category, gender, etc.)
  - Two contestants required 44.5 hours and 55.5 hours.
  - Hollerith's punch card system completed the task in just 5.5 hours.
- His company is a direct predecessor of IBM

Source: [Census.gov: The Hollerith Machine]



The Digital Revolution

ENIAC (1945): Among the first programmable, electronic, general-purpose, digital computers

- Time-sharing systems on mainframes

Theoretical breakthroughs:

- Relational database model (Codd, 1970)
- SQL standard enables data querying at scale

Source: By Unknown author - U.S. Army Photo, Public Domain, https://commons.wikimedia.org/w/index.php?curid=55124;

ENIAC | Penn Engineering

New York Times: Feb. 15, 1946

Electronic computing solved the tabulation problem …

… but created exponentially more data.


Continued on Page 14, Column 6 | Continued on Page 3, Column 2

Electronic Computer Flashes Answers, May Speed Engineering

By T. R. KENNEDY Jr.
Special to THE NEW YORK TIMES.

PHILADELPHIA, Feb. 14—One of the war's top secrets, an amazing machine which applies electronic speeds for the first time to mathematical tasks hitherto too difficult and cumbersome for solution, was announced here tonight by the War Department. Leaders who saw the device in action for the first time heralded it as a tool with which to begin to rebuild scientific affairs on new foundations.

Such instruments, it was said, could revolutionize modern engineering, bring on a new epoch of industrial design, and eventually eliminate much slow and costly trial-and-error development work now deemed necessary in the fashioning of intricate machines. Heretofore, sheer mathematical difficulties have often forced designers to accept inferior solutions of their problems, with higher costs and slower progress.

The "Eniac," as the new electronic speed marvel is known, virtually eliminates time in doing such jobs. Its inventors say it computes a mathematical problem 1,000 times faster than it has ever been done before.

The machine is being used on a problem in nuclear physics.

The Eniac, known more formally as "the electronic numerical integrator and computer," has not a single moving mechanical part. Nothing inside its 18,000 vacuum tubes and several miles of wiring moves except the tiniest elements of matter-electrons. There are, however, mechanical devices associated with it which translate or "interpret" the mathematical language of man to terms understood by the Eniac, and vice versa.

Ceremonies dedicating the machine will be held tomorrow night at a dinner given a group of Government and scientific men at the University of Pennsylvania, after

Column 3

Source: TimesMachine: February 15, 1946 - NYTimes.com

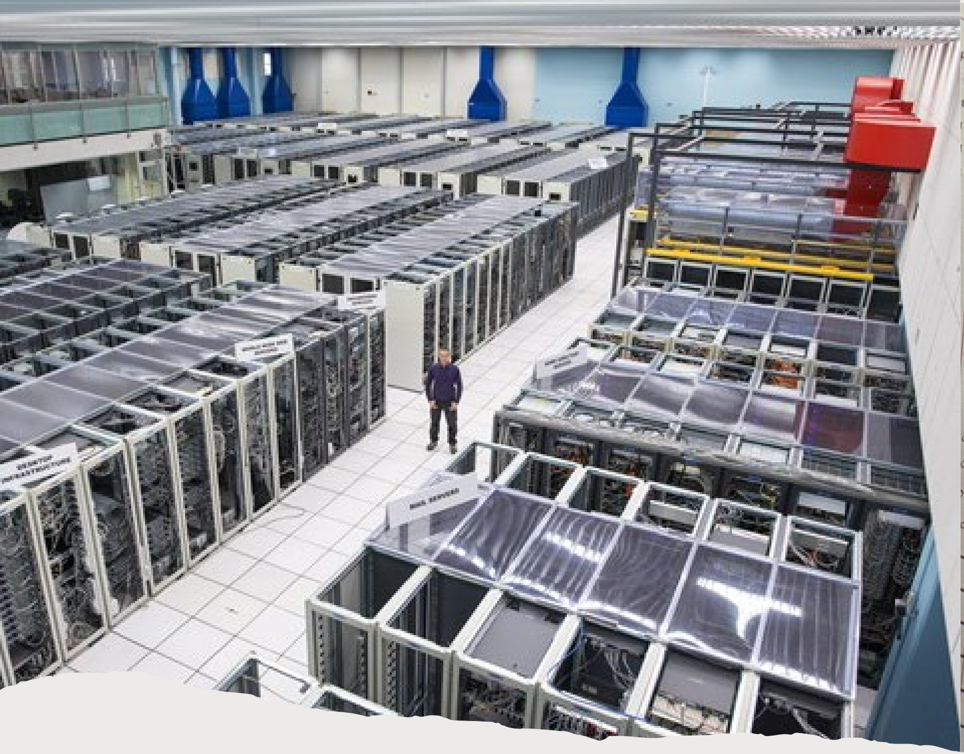
The Network Explosion

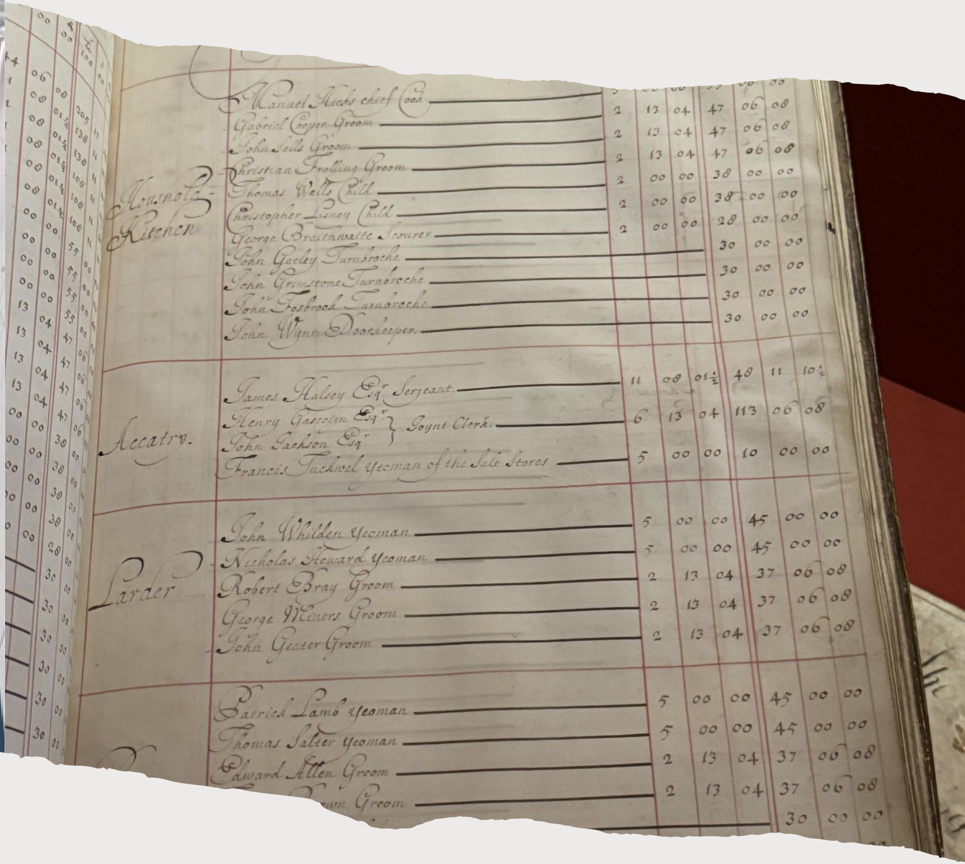
1986: NSFNET connects supercomputer centers (56 Kbps)
1991: The launch of the World Wide Web in 1991 by Tim Berners-Lee and colleagues at CERN; NSFNET at 45 Mbps
1993: 2+ million computers connected; public access mandated
1998: Privatization of internet backbone

Connectivity solved access, but every connection generates data.

The scale fundamentally changed.

Source: *NSFNET Logo Author: Merit Network, Inc. Source: Merit Network, Inc. URL: http://www.merit.edu/news/newsarchive/articles/images/nsfnet_logo.jpg*





Household Kitchen	Manuel Hicks chief Cook	2	13	04	47	06	08
	Gabriel Cooper Groom	2	13	04	47	06	08
	John Sells Groom	2	13	04	47	06	08
	Christian Frolling Groom	2	00	00	38	00	00
	Thomas Wells Child	2	00	00	38	00	00
	Christopher Lisney Child	2	00	00	28	00	00
	George Brastnwaite Scourer				30	00	00
	John Gatley Turnbroche				30	00	00
	John Grimstone Turnbroche				30	00	00
	John Fosbrook Turnbroche				30	00	00
	John Wynn Doorkeeper				30	00	00
Accatry.	James Halsey Esqr Serjeant	11	08	01½	48	11	10½
	Henry Gascoin Esqr / John Jackson Esqr Joynt Clerk	6	13	04	113	06	08
	Francis Tuckwel Yeoman of the Salt Stores	5	00	00	10	00	00
Larder	John Whilden Yeoman	5	00	00	45	00	00
	Nicholas Howard Yeoman	5	00	00	45	00	00
	Robert Bray Groom	2	13	04	37	06	08
	George Miners Groom	2	13	04	37	06	08
	John Geater Groom	2	13	04	37	06	08
	Patrick Lamb Yeoman	5	00	00	45	00	00
	Thomas Salter Yeoman	5	00	00	45	00	00
	Edward Allen Groom	2	13	04	37	06	08
	[illegible] Groom	2	13	04	37	06	08
					30	00	00

Same fundamental activity, unrecognizably different scale.

Sources: CERN, Kensington Palace Account Ledger

The Modern Data Explosion

The Current Inflection Point

Why Now?

1) Storage cost collapse:
From dollars per MB to fractions of a cent per GB
2) Compute availability:
Cloud democratizes access to processing power
3) Network bandwidth explosion:
Global connectivity enables real-time data flows

These exponential changes created an inflection point: not just "more data" but a qualitative shift requiring new thinking.

Why Big Data Came from Tech

Big data tools emerged from business necessity, not academic research:

- Google: MapReduce (2004) — indexing the entire web
- Yahoo: Hadoop (2006) — open-source distributed processing
- Amazon: AWS (2006) — excess infrastructure as service
- Facebook: Cassandra, Presto — social graph at scale

Their use cases (web indexing, ads, recommendations) shaped the tools.

Data Roles

You may work with people in (or take on) these roles:

- **Data Engineer:** Builds and maintains pipelines, infrastructure
- **Data Analyst:** Explores data, creates reports, answers business questions
- **Data Scientist:** Builds models, conducts deeper analysis
- **ML Engineer:** Productionizes models, focuses on deployment

Policy professionals aren't usually any of these; but need to work with all of them.

The Last Mile Problem

The hardest part isn't the data or the analysis.

It's translation into action.

- A model that predicts student dropouts is useless without an intervention
- A fraud detection system is useless if investigators can't act on alerts
- A dashboard is useless if decision-makers don't trust it

Technical capability without operational integration is theater.

The Hidden Costs of Big Data

The full cost of big data systems:

- **Infrastructure:** Storage, compute, networking (the obvious part)
- **Data engineering:** Pipeline building and maintenance (often 70%+ of effort)
- **Quality monitoring:** Ongoing, never finished
- **Governance & compliance:** Grows with scale and sensitivity
- **Technical debt:** Accumulates faster in data systems

Practical wisdom: The best data architecture is the simplest one that solves your problem

What Produces Big Data Today?

- Social media platforms
- Financial markets and transactions
- Power grid and utility systems
- Transportation networks
- Search engines and web activity
- IoT sensors (environmental, industrial, consumer)
- Government administrative systems

Defining Big Data

Frameworks and Definitions

The Core Insight

Big data is data that breaks your existing tools.

The threshold varies by organization and capability. For some, it's hundreds of gigabytes. For others, hundreds of terabytes. The concept is relational, not absolute.

The "Three V's" Framework

- **Volume:** Refers to the size of the dataset (terabytes to petabytes)
- **Velocity:** Refers to the rate of data flow; the speed of generation, collection, and required processing (batch to real-time)
- **Variety:** Refers to data from multiple repositories, domains, or types; the diversity of formats and sources (structured, semi-structured, unstructured)

Source: Laney, D. (2001). 3D Data Management: Controlling Data Volume, Velocity and Variety. Application Delivery Strategies, Meta Group

The Fourth V: It Depends Who's Asking

- NIST emphasizes
  - **Variability**: changes in dataset, whether data flow rate, format/structure, semantics, and/or quality that impact the analytics application; data characteristics change over time
  - The systems challenge
- Industry often emphasizes
  - **Veracity:** the accuracy of the data and trustworthiness
  - The decision challenge
- For policy work, both matter
  - BUT: Veracity deserves special attention because decisions have consequences

Source: NIST Big Data Interoperability Framework

Other V's (Reference)

- **Validity:** Refers to appropriateness of the data for its intended use
- **Value:** Refers to the inherent wealth, economic and social, embedded in any dataset; usefulness and actionable insights
- **Volatility:** Refers to the tendency for data structures to change over time; how long data remains relevant
- The proliferation is partly marketing; three or four V's capture the core insight

Source: NIST Big Data Interoperability Framework

Industry Definition: AWS

“

Big data can be described in terms of data management challenges that – due to increasing volume, velocity and variety of data – cannot be solved with traditional databases. While there are plenty of definitions for big data, most of them include the concept of what’s commonly known as “three V’s” of big data:

Volume: Ranges from terabytes to petabytes of data

Variety: Includes data from a wide range of sources and formats (e.g. web logs, social media interactions, ecommerce and online transactions, financial transactions, etc.

Velocity: Increasingly, businesses have stringent requirements from the time data is generated, to the time actionable insights are delivered to the users. Therefore, data needs to be collected, stored, processed, and analyzed within relatively short windows – ranging from daily to real-time

”

Source: AWS: “What is Big Data?”

Industry Definition: Azure

“

A big data architecture is designed to handle the ingestion, processing, and analysis of data that is too large or complex for traditional database systems.
The threshold at which organizations enter into the big data realm differs, depending on the capabilities of the users and their tools. For some, it can mean hundreds of gigabytes of data, while for others it means hundreds of terabytes.
As tools for working with big datasets advance, so does the meaning of big data. More and more, this term relates to the value you can extract from your data sets through advanced analytics, rather than strictly the size of the data, although in these cases they tend to be quite large.

”

Source: [Azure: “Big Data Architectures”]

Government Definition: NIST

“

Big Data consists of extensive datasets—primarily in the characteristics of volume, variety, velocity, and/or variability — that require a scalable architecture for efficient storage, manipulation, and analysis.

”

Source: NIST Big Data Interoperability Framework

A Synthesis: The Big Data Proposition

Big data is not just lots of data—it is a fundamental shift that concerns:

1. The characteristics of the datasets
2. The architectures of the systems that handle them
3. The methods of analysis
4. Considerations of cost-effectiveness

Source: [NIST Big Data Interoperability Framework]

Data Provenance

The Recorder’s Lens

The Inversion of the Scientific Method

Traditional Research:
Question → Hypothesis → Data Collection → Analysis → Conclusion

Big Data Research:
Data Exists → Explore → Discover Patterns → Generate Hypotheses

This is a genuine epistemological shift with real consequences.

Correlation at Scale

Big data is exceptionally good at finding correlations.

Big data is exceptionally bad at establishing causation.

Policy decisions typically require causal claims and causal methods.

The tension:
Big data can tell you that something is happening, but it often can't tell you why. (Or whether your intervention will change the results.)

Natural experiments can be found and causal methods applied, but industry is only recently catching up to research.

Recording is never neutral

- Viking sea kings employed skalds (poets) to compose sagas celebrating their deeds.
- The sagas we have reflect what powerful men wanted remembered.

- Chinese dynastic records document their nomadic adversaries (Xiongnu, early Türks) in detail, but from the Chinese perspective.
- The nomads' own oral traditions were rarely written down.

History is written by the literate, not necessarily the victors.

Every dataset has provenance:
- Who collected it, and why?
- What didn't get recorded?
- What were the incentives of the collectors?
- How has the collection process changed over time?

The Recorder's Lens

We see through the recorder's lens:

- **Ancient:** Skalds recorded what jarls paid them to record
- **Medieval:** Chinese bureaucrats recorded what mattered to the Chinese state
- **Colonial:** European administrators recorded what enabled extraction and control
- **Modern:** Platforms record what enables advertising and engagement

The pattern persists: recording serves the recorder's purposes

- Big data is usually not gathered specifically to answer the question being asked.
- Understanding the data generating process is vital to correct for any inherent bias.

Designed Data vs. Data Exhaust

The data's history shapes what questions you can legitimately answer.

Designed Data:
- Collected intentionally for analysis
- Surveys, experiments, purpose-built sensors
- You control what gets measured

Data Exhaust:
- Generated as byproduct of operations
- Transaction logs, clickstreams, administrative records
- Cheap, continuous, captures real behavior—but wasn't designed for your question

Big data is largely exhaust.

Questions to Ask

Before you analyze any dataset:

1. Who created this data, and for what purpose?
2. What's missing? Who or what is systematically excluded?
3. How were categories and definitions determined?
4. Has the collection process changed over time?
5. What would make this data misleading for my question?

Big Data Analysis in Action

NYC 311 as an Example

Example: NYC 311 - A Policy Data Laboratory

Since 2010, NYC's 311 system has received over 30 million service requests. During 2014-2023, volume increased 50%. In 2020 alone, 3.23 million requests were logged during the COVID-19 pandemic.

Applying the V's Framework:

- Volume: 30+ million records; 2-3 million annually; grows daily
- Velocity: Real-time submissions via phone, web, mobile; agencies need rapid response
- Variety: 210 complaint types; structured fields + unstructured text + geospatial

Source: NYC Open Data (data.cityofnewyork.us); "311 Service Requests from 2010 to Present"

NYC 311: Variability in Action

Variability: The data changes over time

New channels added: Phone (2003) → Web/Mobile (2009) → MTA integration (2021)
New complaint types emerge: COVID-related categories appeared in 2020
Volume patterns shift: Record 348,463 requests in August 2020
Quality issues: Invalid zip codes, date entry errors, inconsistent categorization

If you're comparing 2012 to 2022, are you comparing like to like?

Source: Chen et al. (2025), "Principles for Open Data Curation: A Case Study with the NYC 311 Service Request Data," arXiv:2502.08649

Example: NYC 311 - Veracity and the Recorder's Lens

Veracity: Can you trust this data for your question?

311 data reflects who reports, not just what problems exist.

Reporting bias: "Groups that are more civically engaged tend to skew whiter and more affluent."
Under-reporting: Lower-income and minority neighborhoods less likely to report street conditions
Gentrification effect: Per capita noise complaints increased 70% in gentrifying neighborhoods
Duplicate reports: Same issue reported multiple times; serial complainers exist

Who's doing the recording shapes what gets recorded.

Source: Hunter Urban Review, "The Value of 311 Data" (2022)

NYC 311: Policy Questions This Enables

Despite the limitations, 311 data enables important questions:

- Can we predict infrastructure failures before they happen?
- Can we identify service inequities across neighborhoods?
- How do response times vary by complaint type and location?
- What does complaint volume tell us about neighborhood change?

The key: Account for the bias, don't ignore it.

Community groups use 311 strategically—increased complaint volume demands official response, even if individual complaints don't.

NYC 311 References

Primary Data:
•NYC Open Data Portal: data.cityofnewyork.us
•Dataset: "311 Service Requests from 2010 to Present" (erm2-nwe9)
•API Access: Socrata Open Data API (SODA)

Academic Research on Reporting Bias:
•Minkoff, S. (2016). "NYC 311: A Tract-Level Analysis of Citizen-Government Contacting." Urban Affairs Review, 52(2): 211-246.
•Kontokosta, C. et al. (2021). "Bias in Smart City Governance: How Socio-Spatial Disparities in 311 Complaint Behavior Impact Fairness." Sustainable Cities and Society.
•Legewie, J. & Schaeffer, M. (2016). "Contested Boundaries: Explaining Where Ethnoracial Diversity Provokes Neighborhood Conflict." American Journal of Sociology, 122(1): 125-161.
•Chen et al. (2025). "Principles for Open Data Curation: A Case Study with NYC 311." arXiv:2502.08649.

Journalism and Policy Briefs:
•Vo, L.T. (2018). "They Played Dominoes Outside Their Apartment for Decades. Then the White People Moved In." BuzzFeed News.
•Data Collaborative for Justice (2019). "Understanding New York City's 311 Data." John Jay College.
•Hunter Urban Review (2022). "The Value of 311 Data."
•Johnson, S. (2010). "What a Hundred Million Calls to 311 Reveal About New York." WIRED.

Types of Data

A General Taxonomy

Structured Data

- Organized into rows and columns
- Conforms to a pre-defined schema
- Relational databases, SQL
- **Examples:**
    - CRM systems
    - Financial databases
    - Inventory management

Semi-Structured Data

- Has organizational properties but not rigid schema
  - Overarching structure, but individual elements are unstructured
- Key-value stores, JSON, XML
- NoSQL databases
- Examples:
  - Web logs
  - API responses
  - E-commerce product catalogs

Unstructured Data

- No predefined data model
- Requires specialized storage and analytics
- BLOBs (Binary Large Objects)
- **Examples:**
  - Audio
  - Video
  - Images
  - Free-text documents
  - Social media posts

Small Vs. Big Data

Small VS Big Data Issues

	Small Data is usually...	Big Data is usually...
Purpose	Gathered for a specific,known goal	May or may not be gathered for analytical purpose
Location	One computer / file	Many systems / files
Structure	Tabular or Excel	May have multiple formats
Preparation	End user for their own goals	Prepared and used by separate people
Collection	Often collected with a single method	May incorporate multiple incorporation methods
Governance	Single source	May involve multiple sources, each with own special rules

Preview: Scaling for Big Data

Architectural Approaches

Vertical Scaling (Scaling Up)

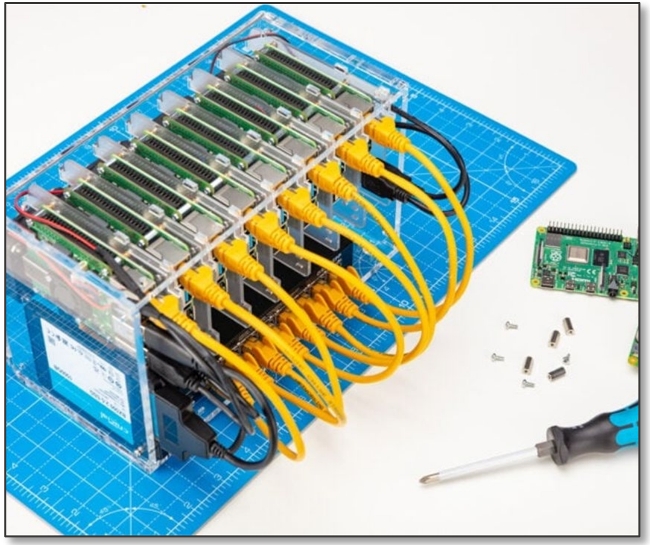
- When you need a bigger boat
- Get a “bigger” machine
  - (More CPU, GPU, RAM, Storage, etc.)
- Expensive; hits hardware limits
- Single point of failure
- Simpler to manage

*“**Vertical scaling** (aka optimization) is the activity to increase data processing performance through improvements to algorithms, processors, memory, storage, or connectivity.”*

Source: [NIST Big Data Interoperability Framework]

Horizontal Scaling (Scaling Out)

- Add more machines to the system
- Can scale indefinitely (in theory)
- Requires load balancing
- Data consistency management
- This is how modern big data systems work



*“**Horizontal scaling** is increasing the capacity of production output through the addition of contributing resources.”*

Source: [NIST Big Data Interoperability Framework]
[How to build a Raspberry Pi cluster - Raspberry Pi]

The Cloud Solution

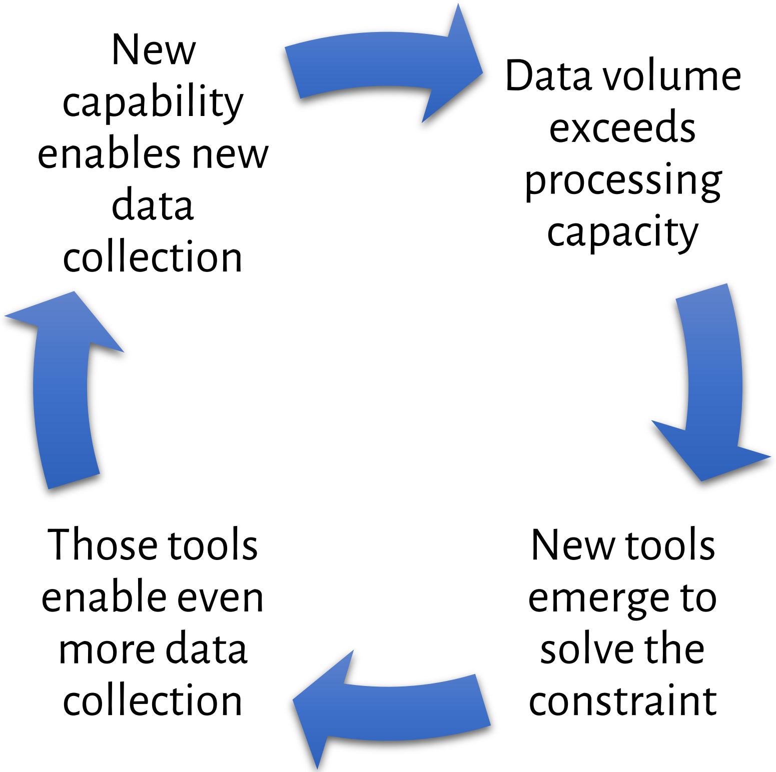
- Outsource infrastructure to specialized providers
- Scale vertically *and* horizontally on demand
- Access enterprise-grade tools at competitive cost
- **Next week:** Deep dive into cloud computing fundamentals

Source: [The Ashburn Data Center Building Boom Hits New Heights | Data Center Frontier]

Summary

Key Lessons

From cuneiform to CERN, we see the same pattern:



This course teaches you the tools for the current iteration of this cycle.

Three Things Big Data Can't Do

"The Algorithm Is Objective"

- Algorithms encode human decisions: what to collect, how to label, what to optimize
- "The algorithm decided" = "humans made choices, then automated them"

"Real-Time Is Always Better"

- Real-time is expensive and complex
- Match processing speed to decision speed

"We Just Need More Data"

- If your data has systematic bias, more data doesn't fix it
- You just get more confident in a wrong answer

Takeaways

Takeaway 1: Big data is relational, not absolute.
"Big" is defined by what breaks your current tools. The threshold differs by organization, era, and capability.
Takeaway 2: The constraint shifts, but the pattern repeats.
Every era generates more data than it can process. You're learning the tools for the current iteration.
Takeaway 3: Recording and analysis are distinct challenges.
We've often been better at gathering data than making sense of it. Modern big data sometimes inverts the traditional sequence: capture everything, discover questions later.
Takeaway 4: More data ≠ better decisions.
Veracity matters. Scale amplifies both signal and noise. Critical evaluation of data quality is not optional.

Next Week

- Structured Data
- SQL
- Relational Databases

Questions?

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